PV101 CTS Cohort 6 Day 2 Master Print Packet

Electricity Basics, Multimeter Practice, Series and Parallel

Program	Cohort	Day	Date
PV101	CTS Cohort 6	Day 2	2026-07-07

Source control note: This packet is intentionally titled and dated for reuse. Do not leave generated curriculum unlabeled in chat history, because apparently civilization enjoys preventable chaos.

Architecture of the Day

- Theme: Solar is not magic. It is light becoming electrical behavior that we can measure, predict, and respect.
- Core idea: Before students install parts, they need to understand what the system is trying to do.
- Primary skill: Explain voltage, current, resistance, power, AC/DC, and basic series/parallel behavior using plain language.
- Assessment: Meter practice sheet, quiz, labeled circuit/system sketch, and exit ticket.
- Room tone: Calm, practical, low-drama, one task at a time.

Learning Targets

- Students explain voltage, current, resistance, and power in simple language.
- Students identify the difference between AC and DC.
- Students use a multimeter to safely measure DC voltage.
- Students explain why PV modules can be energized when exposed to light.
- Students predict what changes in series and parallel connections.
- Students connect the basics to the Cohort 6 RV solar capstone.

Materials

- Digital multimeters with leads
- AA, AAA, and 9V batteries
- Small 12V battery or low-voltage training power supply
- Small PV training panel or module
- Low-voltage lamp, LED, or fan load
- Switch, fuse holder, terminal strip, jumper wires
- Whiteboard markers, pencils, student packets, quiz copies
- PPE as appropriate for the work area

Instructor Day Flow

Time	Block	Instructor Action
10:00-10:15	Arrival reset	Green/yellow/red check-in. Set a calm safety tone. Board: Solar work rewards patience. Guessing is not a method.
10:15-10:35	Day 1 review	PV = photovoltaic. Photo = light. Voltaic = electricity/voltage. Connect review to the RV solar capstone.
10:35-11:05	Electricity basics	Teach voltage, current, resistance, power, and Watts = Volts x Amps. Keep it concrete and example-based.
11:05-11:20	AC vs DC	Solar panels and batteries are DC. Homes and many appliances are AC. Inverters translate DC to AC.
11:20-11:30	Break	Move bodies. Reset attention.
11:30-12:05	Multimeter practice	Inspect leads, set meter, measure batteries and low-voltage sources. Do not freestyle current measurement.
12:05-12:30	PV light demo	Measure panel voltage in light, shade, tilted, and covered conditions. Have students predict first.
12:30-1:00	Lunch	Clean table reset before leaving.
1:00-1:35	Series and parallel	Series stacks voltage. Parallel stacks amperage. Use math examples.
1:35-2:10	Mini-build or circuit trace	Source -> fuse -> switch -> load -> return. Students label current path and test points.
2:10-2:25	RV capstone bridge	Start the load list. AC/DC, voltage, watts/amps, hours per day, notes.
2:25-2:45	Quiz	10-question Day 2 quiz.
2:45-2:55	Exit ticket	Understanding, practice need, safety rule, RV load to verify.
2:55-3:00	Cleanup	Meters off, leads wrapped, tools returned, batteries stored, papers collected.

Board Notes / Anchor Language

Term	Plain Meaning
PV	Photovoltaic: photo = light, voltaic = electricity/voltage.
Voltage	Electrical pressure. Unit: volts (V).
Current	Electrical flow. Unit: amps (A).
Resistance	Opposition to flow. Unit: ohms.
Power	Work being done. Unit: watts (W).
Formula	Watts = Volts x Amps
DC	Direct current. One direction. Solar panels and batteries.
AC	Alternating current. Changes direction. Wall outlets and many appliances.
Series	Voltage adds. Current stays about the same.
Parallel	Current adds. Voltage stays about the same.

Multimeter Safety Rules

1. Inspect the meter and leads before use.
2. Set the meter before touching the circuit.
3. Black lead goes in COM.
4. Red lead goes in the correct port for the measurement.
5. Voltage is measured across two points.
6. Current is measured in series, not across a battery.
7. Never short a battery with meter leads.
8. If unsure, stop and ask.

Instructor Scripts

Moment	Script
Opening	Today we are learning the electrical alphabet of solar: voltage, current, resistance, power, AC/DC, and how panels behave when wired together.
Meter demo	The meter only does what you told it to do. If you tell it something wrong, it may obey. Set the meter before touching the circuit.
Series/parallel	Series stacks voltage. Parallel stacks amperage. We choose based on the system, not vibes.
RV bridge	The RV may contain mystery wiring and hidden fuses. Our rule is verification over labels.
Closing	Today was the alphabet. Tomorrow we start making system sentences: loads, batteries, controllers, inverters, wire, and fuses.

Student Handout and Practice Sheet

Big Idea

- Solar PV turns light into DC electricity. We use measurements instead of guesses.

Electricity Words

- Voltage = pressure
- Current = flow
- Resistance = opposition
- Power = work being done

Units

- Voltage = volts = V
- Current = amps = A
- Resistance = ohms
- Power = watts = W

Main Formula

- Watts = Volts x Amps
- Amps = Watts / Volts
- Volts = Watts / Amps

AC vs DC

- DC = direct current: batteries, solar panels, many RV loads
- AC = alternating current: wall outlets and many home appliances

Series and Parallel

- Series = voltage adds
- Parallel = current adds
- Memory line: Series stacks voltage. Parallel stacks amperage.

PV System Path

- Solar panel -> charge controller -> battery -> loads
- Battery -> inverter -> AC loads when needed

Meter Practice Record

Object Tested	Meter Setting	Expected Reading	Actual Reading	What Did It Tell You?
AA battery	DC volts			
9V battery	DC volts			
12V source	DC volts			
Small PV panel in light	DC volts			
Small PV panel shaded/covered	DC volts			
Wire continuity	Continuity			

RV Solar Load Starter List

Load	AC/DC	Voltage	Watts or Amps	Notes
Water pump	DC	12V	Unknown	Verify hidden fuse and label
Interior lights	DC	12V	Unknown	Count fixtures
Fan	DC	12V	Unknown	Check label or measure
USB charging	DC	5V/12V	Unknown	Adapter losses
Outlet loads	AC	120V	Unknown	Inverter needed

Exit Ticket

One thing I understand better today:

_____

One thing I still want to practice:

_____

One safety rule I need to remember:

_____

One RV load we should verify before designing the system:

_____

PV101 Day 2 Quiz

Name: _____ Date: 7/7/2026

1. What does PV stand for?

- A. Power Voltage
- B. Photovoltaic
- C. Panel Verification
- D. Positive Value

Answer: _____

2. Solar panels produce what kind of electricity?

- A. AC
- B. DC
- C. Static only
- D. None unless plugged into a wall

Answer: _____

3. Match: Voltage, Current, Resistance, Power.

- Voltage = electrical pressure
- Current = electrical flow
- Resistance = opposition to flow
- Power = work being done

Answer: _____

4. What formula gives watts?

- A. Watts = Volts x Amps
- B. Watts = Ohms / Volts
- C. Watts = Amps + Volts
- D. Watts = Resistance x Time

Answer: _____

5. A 12V fan draws 3A. How many watts is it using?

Work: _____

Answer: _____

6. A 60W light is running on a 12V system. How many amps does it draw?

Work: _____

Answer: _____

7. In series wiring, what usually increases?

- A. Voltage
- B. Resistance only
- C. Wire color
- D. Nothing

Answer: _____

8. In parallel wiring, what usually increases?

- A. Voltage
- B. Current
- C. Sunlight
- D. Panel size

Answer: _____

9. True or false: A solar panel can produce voltage whenever light is hitting it.

Answer: _____

10. Why should you inspect meter leads before using a multimeter?

Answer: _____

PV101 Day 2 Quiz Answer Key

1. What does PV stand for?
Answer: B

2. Solar panels produce what kind of electricity?
Answer: B

3. Match: Voltage, Current, Resistance, Power.
Answer: Voltage=pressure; Current=flow; Resistance=opposition; Power=work

4. What formula gives watts?
Answer: A

5. A 12V fan draws 3A. How many watts is it using?
Answer: 36W

6. A 60W light is running on a 12V system. How many amps does it draw?
Answer: 5A

7. In series wiring, what usually increases?
Answer: A

8. In parallel wiring, what usually increases?
Answer: B

9. True or false: A solar panel can produce voltage whenever light is hitting it.
Answer: True

10. Why should you inspect meter leads before using a multimeter?
Answer: Accept: damaged leads can give bad readings, short risk, or unsafe testing.

Google Classroom Post

PV101 Day 2 - Electricity Basics + Meter Practice

Today we covered the foundation needed for solar work: voltage, current, resistance, watts, AC vs DC, safe multimeter use, and the difference between series and parallel wiring.

Main idea:

Solar panels produce DC electricity when exposed to light.

Voltage pushes.

Current flows.

Resistance limits.

Power does work.

Watts = Volts x Amps.

Hands-on work:

- Measured batteries and low-voltage sources with a multimeter

- Observed how light and shade affect a small solar panel

- Compared series and parallel wiring

- Started connecting electrical basics to the RV solar capstone

Exit ticket:

1. One thing I understand better today

2. One thing I still want to practice

3. One safety rule I need to remember

4. One RV load we should verify before designing the system